

* Number System :-	Base/Radix -
• Decimal - 0-9	10
• Binary - 0,1	2
• Ternary - 0,1,2	3
• Octal - 0-7	8
• Hexadecimal - 0-9, A, B, C, D, E, F	16

Eg :- $(512)_{10} = 5 \times 10^2 + 1 \times 10^1 + 2 \times 10^0$

$(1011)_2 = 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$
 $= 8 + 0 + 2 + 1$

$= (11)_{10}$

$(A1.1)_{16} = 10 \times 16^1 + 1 \times 16^0 + 1 \times 16^{-1}$

$(42)_{10} \leftrightarrow 2$

$42 \div 2 = 21$

$21 \div 2 = 10$

$10 \div 2 = 5$

$5 \div 2 = 2$

$2 \div 2 = 1$

$1 \div 2 = 0$

$0 \div 2 = 0$

$= (101010)_2$

$8 \div 2 = 4$

$4 \div 2 = 2$

$2 \div 2 = 1$

$16 \div 2 = 8$

$8 \div 2 = 4$

$4 \div 2 = 2$

$2 \div 2 = 1$

$(0.625)_{10}$

$0.625 \times 2 = 1.25 = 1 + 0.25 = (0.101)_2$

$0.25 \times 2 = 0.5 = 0 + 0.5$

$0.5 \times 2 = 1.0 = 1 + 0$

Eg :- $(85.21)_{10}$

85	⇒	2	85
			42
			21
			10
			5
			2
			1
			0
			0

$= (1010101)_{10}$

$0.21 \Rightarrow 0.21 \times 2 = 0.42$

$0.42 \times 2 = 0.84$

$0.84 \times 2 = 1.68$

$0.68 \times 2 = 0.36 + 1$

Hence, $(1010101.0011)_2$

85	⇒	8	85
			5
			2
			1
			0

$= (125)_8$

$0.21 \times 8 = 1.68$

$0.68 \times 8 = 5.44$

$0.44 \times 8 = 3.52$

$(125.153)_8$

$85 \Rightarrow 16$

$5(A5) = (55)_{16}$

$0.21 \times 16 = 3.36$

$(55.35C)_{16}$

$$(52520)_{10} \rightarrow 10^5 - 52520 = 47480$$

$$(0.3261)_{10} \rightarrow 10^0 - 0.3261 = 0.6733$$

$$(25.36)_{10} \rightarrow 10^2 - 25.36 = 74.64$$

$$(101100)_2 \rightarrow (2^6 - 101100)_{10} = 1000000 - 101100 = 898900$$

$$(101100)_2 \rightarrow (2^6 - 101100)_{10} = 1000000 - 101100 = 898900$$

$$(10011)_2 \rightarrow (2^5 - 10011)_{10} = 100000 - 10011 = 89989$$

$$(0.0110)_2 \rightarrow (2^2 - 0.0110)_{10} = 100 - 0.0110 = 99.989$$

→ (x-1)'s complement (N_c)

'n' → integer part
'm' → fraction part

$$(x-1)'s \rightarrow x^n - x^{-m} = N$$

$$(52520)_{10} \rightarrow 10^5 - 52520 = 47480$$

$$99999 - 52520 = 47479$$

$$(25.639)_{10} \rightarrow (10^2 - 10^{-3}) - 25.639 = 99.999 - 25.639 = 74.360$$

$$(101100)_2 \rightarrow (2^6 - 2^0) - 101100 = (1000000 - 1) - 101100 = 111111 - 101100 = 010011$$

$$(100.001)_2 \rightarrow (011.110)_2 \rightarrow 1's \text{ complement} \rightarrow 11.111$$

• Subtraction with x's complement -

- (1) Make the number of digits in both M and N to be same.
- (2) Find x's complement of N.
- (3) Add M and x's complement of N.
- (4) If you generate carry, discard it. If no carry is generated, take x's complement of the number obtained and place a negative sign in front.

→ Find M - N.

$$\text{Eg :- } M = 72532$$

$$N = 3250 \rightarrow 03250$$

$$\text{For N, } 10's \text{ complement} \rightarrow 10^5 - 03250 = 96750$$

Now, 72532

$$96750$$

$$(discard) 1 \leftarrow 69282 \Rightarrow 69282$$

Eg :- $M = 000100.11000$

$N = 100110.10001$

2's of N $\Rightarrow$

011001.01111

000100.11000

No carry $\leftarrow 011110.00111$

carry

2's $\rightarrow 100001.11001$

Hence, -100001.11001 Ans.

- Subtraction using $(n-1)$'s complement :-

(1) Same

(2) $(n-1)$'s complement of 'N'

(3) M

+ $(n-1)$'s N

(4) If carry is generated, add the carry to LSP.

If carry isn't generated

$M < N \rightarrow$ take $(n-1)$'s complement and put a -ve sign.

$M = 72532$

$N = 03250$

9's of N $\rightarrow 96749$

72532

$1 \leftarrow 69281$

69282

Eg :- 1011.110

$- 0110.110$

1's of N $\rightarrow 1001.001$

$11 \leftarrow 0100.111$

0101.000

2's of N = 1001.010

1011.110

X ← 0101.000

Eg % 00101.101

- 11011.100

1's of N = 00100.011

00101.101

00100.011

No carry ← 01010.000

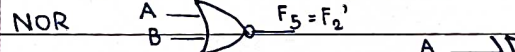
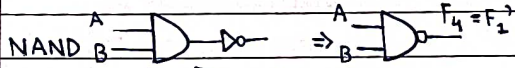
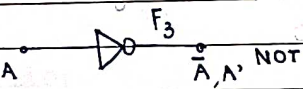
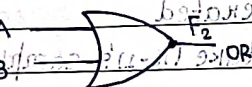
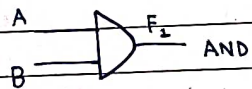
⇒ -10101.111

* AND

OR

NOT

Logical Gates



Truth Table %

A	B	F ₁	F ₂	F ₃	F ₄	F ₅	F ₆	F ₇
0	0	0	0	1	1	1	0	1
0	1	0	1	0	0	0	1	0
1	0	0	1	0	0	0	1	0
1	1	1	0	1	1	0	0	1

*

BINARY CODES %

Decimal codes %

BCD %

Excess-3 %

0 %

1 %

2 %

3 %

4 %

5 %

6 %

7 %

8 %

9 %

complements of reflection (reflective property)

Reflective/self-complementary codes

(42)₁₀ = (101010)₂

BCD → 01000010

Ex-3 → 01110101

flip-bits

10001010

*

Gates % AND = F₁ = A · B

OR = F₂ = A + B

NOT = F₃ = A' = $\overline{A}$

NAND = F₄ = $\overline{A \cdot B}$

NOR = F₅ = $\overline{A + B}$

XOR = F₆ = $\overline{A \cdot B} + \overline{A \cdot B} = A \oplus B$

XNOR = F₇ = $\overline{A \oplus B} = (A \oplus B)'$

(A · B)'

Maxterms

Canonical SOP form

	A	B	F_2 (OR)
$A'B'$	0	0	m_0 0
$A'B$	0	1	m_1 1 ✓
AB'	1	0	m_2 1 ✓
AB	1	1	m_3 1 ✓

$F_2 = A'B + AB' + AB$
Sum of Products (SOP)
 $F_2 = m_1 + m_2 + m_3$
 $= \sum (1, 2, 3)$

Product of Sum (POS)

	A	B	F_2
$A+B$	0	0	M_0 0 ✓
$A+B'$	0	1	M_1 1
$A'+B$	1	0	M_2 1
$A'+B'$	1	1	M_3 1

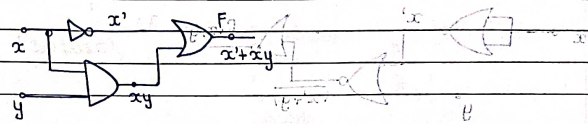
$F_2 = A+B$
 $F_2 = M_0$
 $F_2 = (A+B)(A'+B')$
 $= M_0 M_3$
 $= \prod (0, 3)$

	x	y	z	F_1	F_2
$x'y'z'$	0	0	0	m_0 0	0
$x'y'z$	0	0	1	m_1 1	0
$x'yz'$	0	1	0	m_2 1	0
$x'yz$	0	1	1	m_3 0	1
$xy'z'$	1	0	0	m_4 1	0
$xy'z$	1	0	1	m_5 0	1
xyz'	1	1	0	m_6 0	1
xyz	1	1	1	m_7 1	1

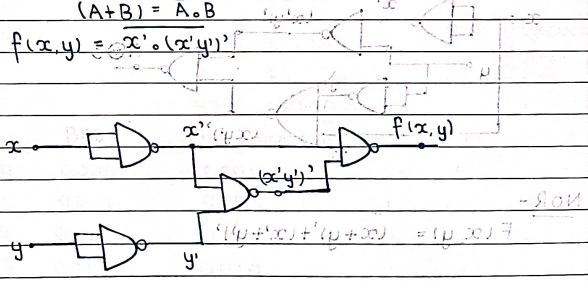
$F_1(x,y,z) = (x+y+z)(x+y'+z')(x'+y+z)$
 $= \prod (0, 3, 5, 6)$

$$F_2(x,y,z) = \prod (0, 1, 2, 4)$$

$$\rightarrow F(x,y) = x' + xy$$



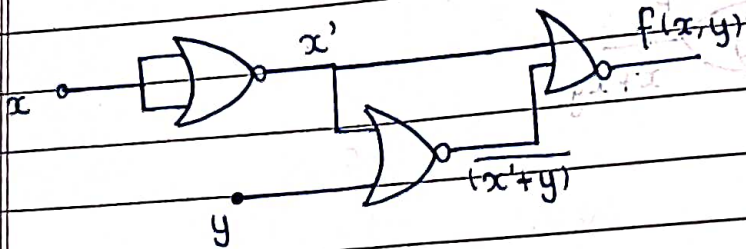
- NAND -
- 1) Represent the fn. in standard SOP form.
 $f(x,y) = x + x'y$
- 2) Take double complement of fn.
 $\Rightarrow f(x,y) = \overline{\overline{f(x,y)}} = \overline{x + x'y}$
- 3) Apply De-Morgan's theorem to internal complement.



- NOR -
- 1) Represent in POS form.
- 2) Double complement
- 3) DeMorgan's theorem
 $\overline{A \cdot B} = \overline{A} + \overline{B}$

$$\overline{f(x,y)} = \overline{x(x'+y)}$$

$$= \overline{x' + (x'y)}$$



Eg :- XOR gate -

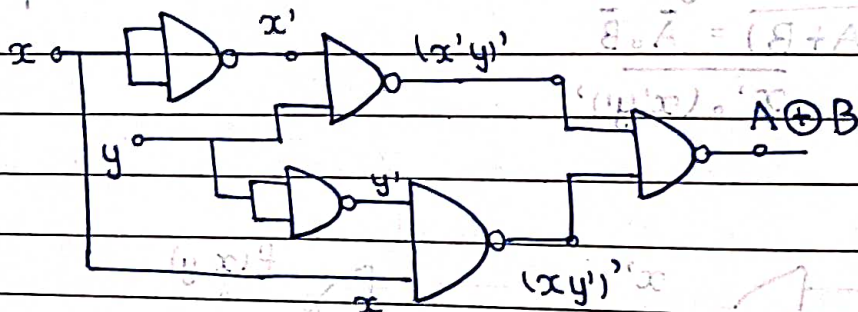
$$F(x,y) = x'y + xy'$$

$$F(x,y) = (x+y)(x'+y')$$

• NAND -

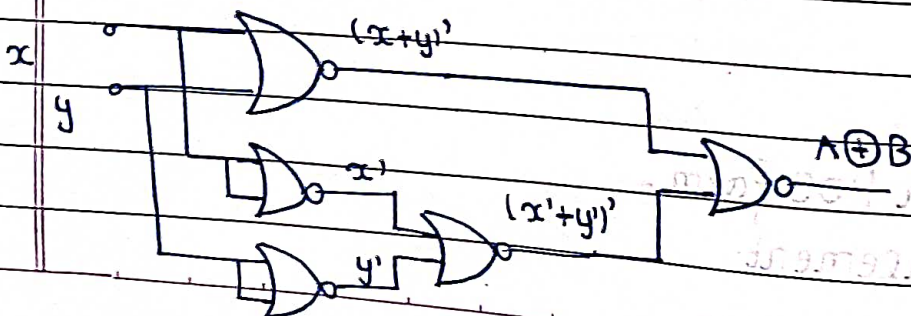
$$F(x,y) = \overline{x'y + xy'}$$

$$= (x'y)'(xy)'$$



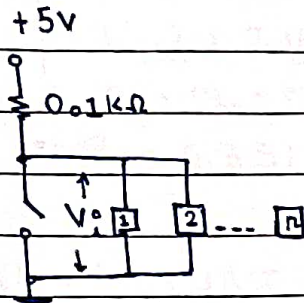
• NOR -

$$F(x,y) = (x+y)' + (x'+y')'$$



iii. Fan Out - No. of similar gates that can be driven by a gate.

Eg :-



If $V_i = 3.5V$, $n = ?$

$$V_i = \frac{2.5}{0.1 + \frac{2}{N}} \times 3.5 \Rightarrow N \approx 8$$

iv. Current and Voltage Parameters - Every -

High lve. input voltage V_{IH}

Low " " " " V_{IL}

High lve. Output " " V_{OH}

etc. V_{OL}
 I_{IH}, I_{IL}

For gate 1 $\rightarrow$ gate 2,

$$V_{OH} > V_{IH} \text{ and } V_{OL} < V_{IL}$$

v. Noise Immunity -

Circuit ability to tolerate noise signals.

In terms of noise margin,

$$1 \text{ State NM} = V_{OH} - V_{IH}$$

$$0 \text{ State NM} = V_{IL} - V_{OL}$$

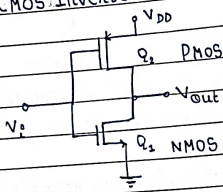
should be high.

vi. Operating Temperature

vii. Power Supply Requirements

viii. Fan-In

* CMOS Inverter



* BINARY LOGIC:-

Boolean Algebra -

AND

OR

NOT

Duality - Every algebraic expression deducible from the postulates of boolean algebra remains valid if operators and identity elements are interchanged.

Operations

Identity Elements

AND

0

OR

1

* Theorem :-

- $x + 0 = x$ $x \cdot 1 = x$
- $x + \bar{x} = 1$ $x \cdot \bar{x} = 0$
- $x + x = x$ $x \cdot x = x$
- $x + 1 = 1$ $x \cdot 0 = 0$
- $(x)' = x$
- Commutative $x + y = y + x$ $x \cdot y = y \cdot x$
- Associative $x + (y + z) = (x + y) + z$ $x \cdot (y \cdot z) = (x \cdot y) \cdot z$
- Distributive $x(y + z) = xy + xz$ $x + yz = (x + y)(x + z)$
- De Morgan's $(x + y)' = x' \cdot y'$ $(x \cdot y)' = x' + y'$
- Absorption $x + xy = x$ $x(x + y) = x$

* Minterms & Maxterms :-

$$F(x, y, z) = x'y + y'z + xy'z$$

$$= x'y(z + z') + (x + x')y'z + xy'z$$

$$= x'yz + x'y'z + xy'z + x'y'z$$

$$= m_3 + m_2 + m_5 + m_1$$

$$= \sum(1, 2, 3, 5)$$

$$F(x, y, z) = (x + y)(x + z)$$

$$= (x + y + z)(x + y + z')$$

$$= (x + y + z)(x + y + z) + (x + y + z)(x + y + z')$$

$$= M_0 M_1 M_2$$

$$= \prod(0, 1, 2)$$

* Simplification :-

$$F(x, y) = x + x'y = (x + x')(x + y)$$

$$= x + y$$

$$F(x, y) = x(x + y) = xy$$

$$F(x, y, z) = x'y'z + x'y'z' + xy'$$

$$= x'z + xy'$$

$$F(a, b, c) = bc + ac' + ab + bcd$$

$$= bc + ac' + ab$$

$$= ab + bc + ac'$$

Eq :- Find maxterms

$$F(a, b, c, d) = (a + b)(b + c + d)$$

Find minterms

$$F(x, y, z) = x'z' + xy' + xyz$$

Ans.

$$F(a, b, c, d) = (a + b + c + d)(a + b + c' + d)$$

$$= (a + b + c + d)(a + b + c' + d)(a + b + c + d)(a + b + c' + d)$$

$$= (a + b + c + d)(a + b + c' + d) = M_0 M_1 M_2 M_3 M_5$$

$$F(x, y, z) = z' + x'y + xy z$$

$$= (x+x')(y+y')z' + x'y(z+z') + xy z$$

$$= xy z' + xy z + x'y z' + x'y z + x'y z + xy z$$

$$= xy z' + xy z + x'y z + x'y z$$

* Karnaugh Map:-

→ Squares

→ Each square represents a minterm.

2 Variable - $F(x, y)$

x \ y	0	1
0	$x'y'$ m_0	$x'y$ m_1
1	xy' m_2	xy m_3

Eg:-

x \ y	0	1
0	1	1
1	1	1

$$x'y' + x'y = x'(y' + y) = x'$$

$$x'y' + xy' = y'(x' + x) = y'$$

3 Variable - $F(x, y, z)$

- Implicant - Any single minterm / permitted group of minterms is called an implicant.
- Prime implicant - Group of minterms which can't be combined with any other minterm / group.
- Essential prime implicant - A prime implicant in which one / more minterms are unique.

x \ yz	00	01	11	10
0	m_0	m_1	m_3	m_2
1	m_4	m_5	m_7	m_6

Eg:-

x \ yz	00	01	11	10
0		1	1	
1	1	1		

$$F = x'y + xy'$$

Eg:-

x \ yz	00	01	11	10
0		1	1	1
1	1	1		

$$F = z + x'y$$

Eg:-

x \ yz	00	01	11	10
0	1	1	1	1
1	1	1	1	1

$$F = yz + x'z' + xz$$

Eg:-

x \ yz	00	01	11	10
0	1			1
1	1	1		1

$$F = z' + xy'$$

1. $F(x, y, z) = \sum(2, 3, 6, 7) = y$

2. $F(a, b, c) = a'b + bc' + b'c' = a'bc' + a'b'c' + a'bc + abc' + abc + abc'$

x \ yz	00	01	11	10
0	1		1	1
1	1			1

$$F = c' + a'b$$

Four Variable- $F(w, x, y, z)$

wxy	00	01	11	10
00	m ₀	m ₁	m ₃	m ₂
01	m ₄	m ₅	m ₇	m ₆
11	m ₁₂	m ₁₃	m ₁₅	m ₁₄
10	m ₈	m ₉	m ₁₁	m ₁₀

$F(w, x, y, z) = \sum(0, 1, 2, 4, 5, 6, 8, 9, 12, 13, 14)$

wxy	00	01	11	10
00	1	1		1
01	1	1		
11	1	1		1
10	1	1		

$F = y' + w'z' + xz'$
(I) (II) (III)

$F(a, b, c, d) = a'b'c'd + a'b'c'd' + ab'cd' + a'b'cd' + a'bcd' + abc'd + abc'd'$
 $\sum(0, 1, 2, 4, 5, 6, 8, 9)$

wxy	00	01	11	10
00	1	1		1
01	1	1		
11	1	1		1
10	1	1		

$= b'c' + b'd' + a'cd'$

Eg:- $a'b'c' \rightarrow$ one variable missing, represent 2 minterms with d and d'

ab	00	01
00	1	1
01	1	1

Eg:- $F(a, b, c, d) = d(a+b) + b'(c+ad)$
 $= a'd + bd + b'c + a'b'd$

ab	00	01	11	10
00		1	1	1
01		1	1	
11		1	1	
10		1	1	1

$F = d + cb'$

$F(a, b, c, d) = abd + a'cd' + a'b + a'cd' + ab'd'$

ab	00	01	11	10
00	1			1
01	1	1	1	1
11		1	1	
10	1			1

$F = bd + b'd' + a'b$
or
 $bd + b'd' + a'd'$

Eg:- $F(w, x, y, z) = (x'z + w'xy' + wx'y' + wxy'z)$

wxy	00	01	11	10
00	1	1		
01	1	1		
11	1	1		
10	1	1	1	1

$F = x'y + x'z + wx'y$

xy	00	01	11	10
00	1	1		
01	1	1		
11	1	1	1	1
10	1	1	1	1

Five Variable Map

$F(a, b, c, d, e) = d(a+b+c) + b'(c+d)$

ab \ cde	000	001	011	010	100	101	111	110
00	0	1	3	2	6	7	5	4
01	8	9	11	10	14	15	13	12
11	24	25	27	26	30	31	29	28
10	16	17	19	18	22	23	21	20

$F(a,b,c,d,e) =$

ab \ cde	000	001	011	010	100	101	111	110
00	1			1	1		1	
01		1	1			1	1	
11		1	1			1	1	
10		1				1		

$F = be + a'be' + ad'e$

* Product of Sum Simplification :-

$F(a,b,c,d) = \sum(0,1,2,5,8,9,10) = \prod(3,4,6,7,11,12,13,14,15)$

ab \ cd	00	01	11	10
00	1	1		1
01		1		
11	1	1		1
10				

$F = b'e' + b'd' + a'c'd$

ab \ cd	00	01	11	10
00	0		0	0
01	0	0	0	0
11	0	0	0	0
10	0		0	0

$F' = cd + ab + bd'$

$(F')' = F = (cd + ab + bd')$

$= (c' + d')(a' + b')(b' + d)$

$F(w,x,y,z) = \prod(0,2,8,10,12,13)$

ab \ cd	00	01	11	10
00	0			0
01	0	0		
11				
10	0			0

$F' = w'd' + abc'$

$F = (b+d)(a'+b'+c)$

ab \ cd	00	01	11	10
00		1	1	
01	1	1	1	1
11			1	1
10		1	1	

$F = a'b' + bca + b'd$

* Don't care Conditions :-

$F(w,x,y,z) = \sum(1,3,7,11,15) + \sum_d(0,2,5)$

wx \ yz	00	01	11	10
00	X	1	1	X
01		X	1	
11			1	
10			1	

$F = yz + w'z$

wx \ yz	00	01	11	10
00	X			X
01	0	X		
11	0	0		
10	0	0		

$F' = z' + wy'$

$F = z(w+y')$

Eg :-

$$F(a,b,c,d) = b'c'd' + bcd' + abcd'$$

don't care = $b'cd' + a'bc'd$

ab \ cd	00	01	11	10
00	1			X
01		X		1
11				1
10	1			X

$F = cd' + b'd'$

* Tabulation Method :-

$$F = \sum (0, 1, 2, 8, 10, 11, 14, 15)$$

wxyz	wxyz	wxyz
0 0000	000 - (0,1)	- 0 - 0 (0,2,8,10)
1 0001	00 - 0 (0,2)	- 0 - 0 (0,8,2,10)
2 0010	- 000 (0,3)	
8 1000	- 010 (2,10)	1 - 1 - (10,11,14,15)
10 1010	- 10 - 0 (8,10)	1 - 1 - (10,14,11,15)
11 1011	101 - (10,11)	
14 1110	1 - 10 (10,14)	
15 1111	1 - 11 (11,15)	
	111 - (14,15)	

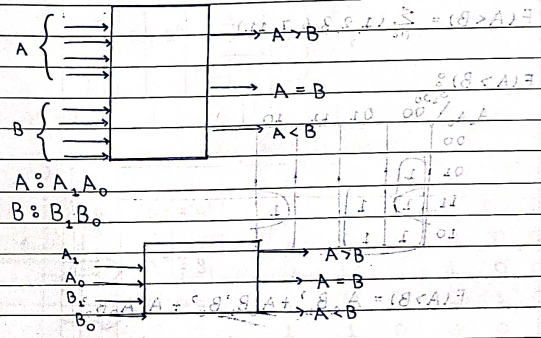
$F = wy + x'z' + w'x'y'$

wy	0	1	2	8	10	11	14	15
$x'z'$	X		(X)	(X)	X		(X)	(X)
$w'x'y'$	X	(X)						

• Magnitude Comparator :- 7485 IC

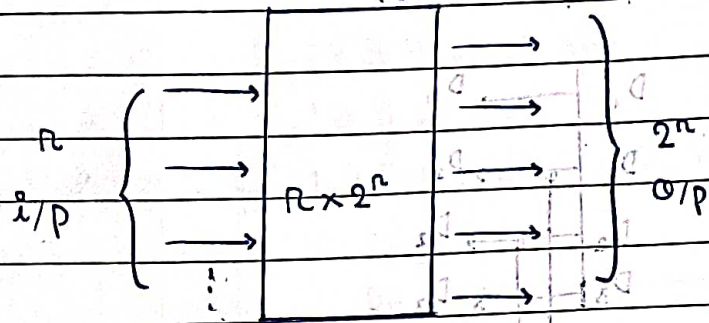
$$A = A_3A_2A_1A_0$$

$$B = B_3B_2B_1B_0$$



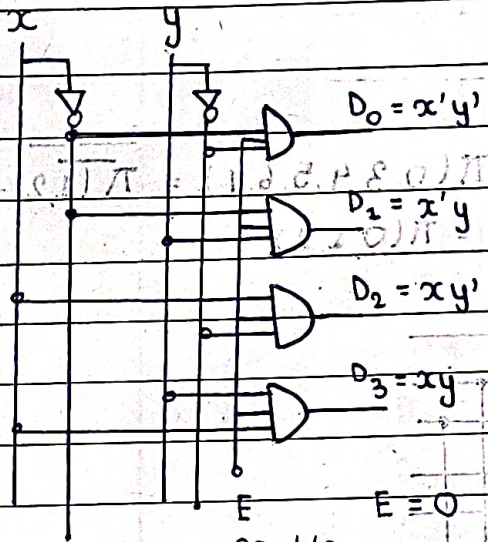
A ₃	A ₂	A ₁	A ₀	B ₃	B ₂	B ₁	B ₀	A > B	A = B	A < B
0	0	0	0	0	0	0	0	0	1	0
0	0	0	0	0	0	1	0	0	0	1
0	0	0	1	0	0	0	0	0	1	0
0	0	1	0	0	0	0	0	0	1	0
0	0	1	1	0	0	0	0	0	1	0
0	1	0	0	0	0	0	0	1	0	0
0	1	0	1	0	0	0	0	1	0	0
0	1	1	0	0	0	0	0	1	0	0
0	1	1	1	0	0	0	0	1	0	0
1	0	0	0	0	0	0	0	0	0	1
1	0	0	1	0	0	0	0	0	0	1
1	0	1	0	0	0	0	0	0	0	1
1	0	1	1	0	0	0	0	0	0	1
1	1	0	0	0	0	0	0	0	0	1
1	1	0	1	0	0	0	0	0	0	1
1	1	1	0	0	0	0	0	0	0	1
1	1	1	1	0	0	0	0	0	0	1

* Decoders :



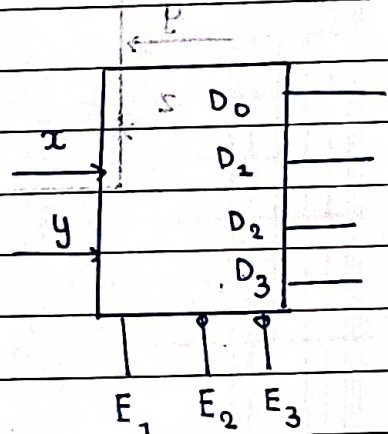
$n = 2, 2^n = 4$

Minterm decoder



x	y	D_0	D_1	D_2	D_3
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1

$E = 0$ disabled
 $E = 1$ enabled



for IC to work

→ Complementary output decoder :-

Using NAND instead of AND gates.

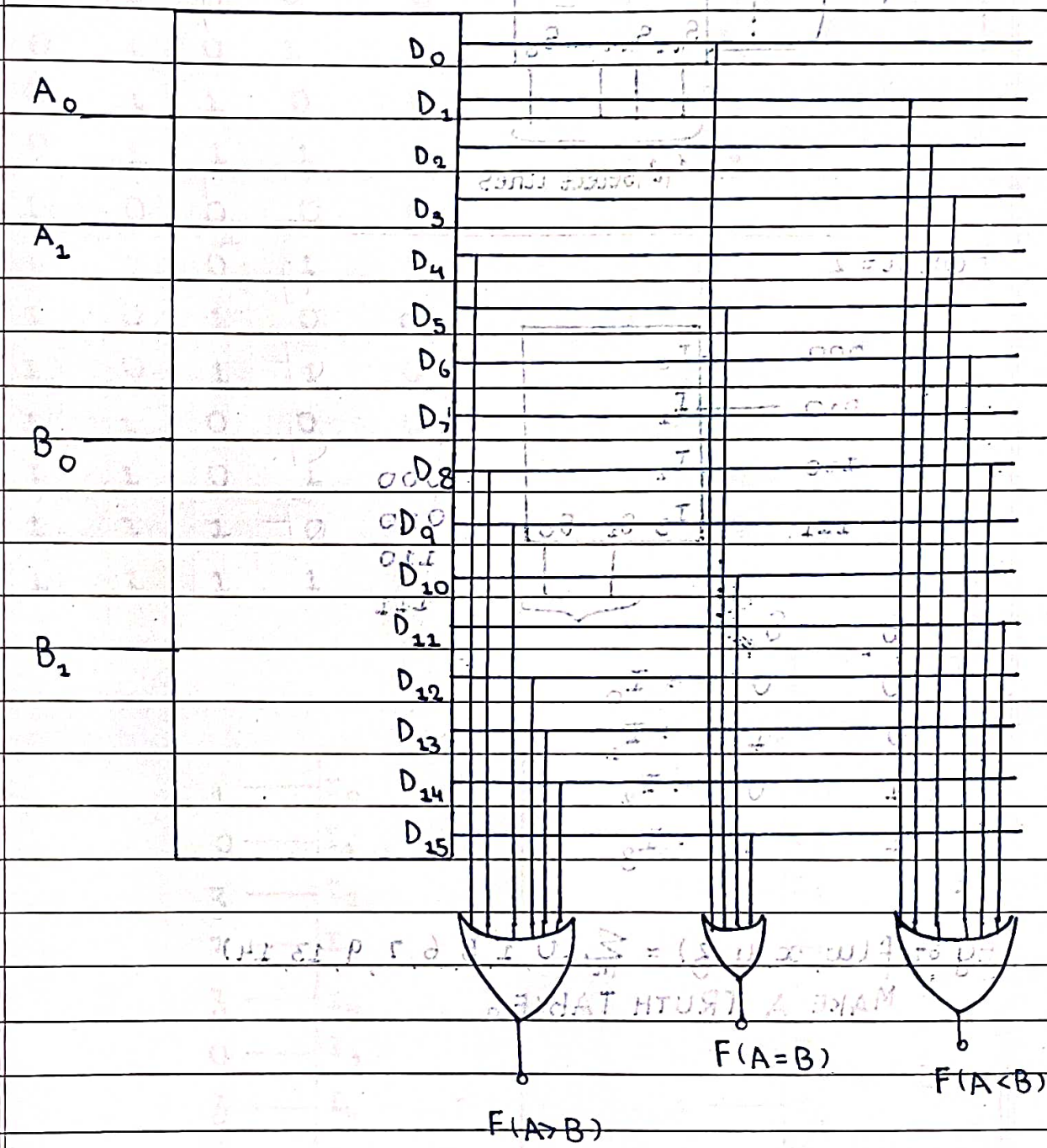
x	y	D_0	D_1	D_2	D_3
0	0	0	1	1	1
0	1	1	0	1	1
1	0	1	1	0	1
1	1	1	1	1	0

• Implementing Magnitude Comparator using decoders -

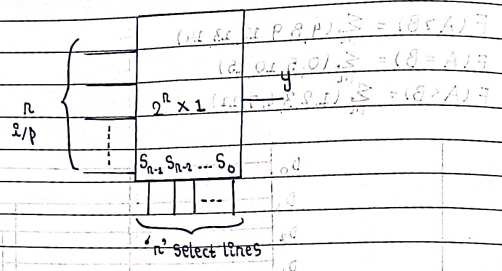
$$F(A > B) = \sum (4, 8, 9, 12, 13, 14)$$

$$F(A = B) = \sum (0, 5, 10, 15)$$

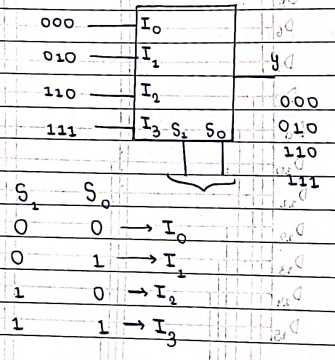
$$F(A < B) = \sum (1, 2, 3, 6, 7, 11)$$



* Multiplexer



For $n=2$,



Eg:- $f(w, x, y, z) = \sum_m (0, 1, 5, 6, 7, 9, 13, 14)$
 MAKE A TRUTH TABLE.

w	x	y	z	f
0	0	0	0	1
0	0	0	1	1
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	0
1	0	0	1	1
1	0	1	0	0
1	0	1	1	0
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0

w	x	y	z	f
1	0	0	0	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

x	y	z	f
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

*

I_0

I_1

I_2

I_3

I_4

I_5

I_6

I_1, S_2, S_1, S_0

w, y, z

$I_0, I_1, I_2, I_3, I_4, I_5, I_6$

δ, ρ, x, w

$I_0, I_1, I_2, I_3, I_4, I_5, I_6$

$I_0, I_1, I_2, I_3, I_4, I_5, I_6$

$I_0, I_1, I_2, I_3, I_4, I_5, I_6$

$I_0, I_1, I_2, I_3, I_4, I_5, I_6$

$I_0, I_1, I_2, I_3, I_4, I_5, I_6$

$I_0, I_1, I_2, I_3, I_4, I_5, I_6$

$I_0, I_1, I_2, I_3, I_4, I_5, I_6$

$I_0, I_1, I_2, I_3, I_4, I_5, I_6$

$I_0, I_1, I_2, I_3, I_4, I_5, I_6$

$I_0, I_1, I_2, I_3, I_4, I_5, I_6$

$I_0, I_1, I_2, I_3, I_4, I_5, I_6$

δ	ρ	x	w
1	0	0	0
1	1	0	0
0	0	1	0
0	1	1	0
0	0	0	1
1	1	0	1
1	0	1	1
1	1	1	0
0	0	0	1
1	1	0	1
0	0	1	1
0	1	1	1
1	1	0	1
1	0	1	1
0	1	1	1

$I_0, I_1, I_2, I_3, I_4, I_5, I_6$

$I_0, I_1, I_2, I_3, I_4, I_5, I_6$

$I_0, I_1, I_2, I_3, I_4, I_5, I_6$

$I_0, I_1, I_2, I_3, I_4, I_5, I_6$

$I_0, I_1, I_2, I_3, I_4, I_5, I_6$

$I_0, I_1, I_2, I_3, I_4, I_5, I_6$

$I_0, I_1, I_2, I_3, I_4, I_5, I_6$

$I_0, I_1, I_2, I_3, I_4, I_5, I_6$

$I_0, I_1, I_2, I_3, I_4, I_5, I_6$

$I_0, I_1, I_2, I_3, I_4, I_5, I_6$

$I_0, I_1, I_2, I_3, I_4, I_5, I_6$